

wireless

WiFi: How it works

The growing popularity of laptops necessitated the need for a wireless standard that would allow people to walk around with their laptops, while still connected to a network. WiFi, an abbreviation for Wireless Fidelity, is a wireless standard that offers mobility while offering data rates comparable to those of a wired LAN. It's actually a collection of standards ratified by the Institute of Electrical and Electronic Engineers (IEEE). In 1997, IEEE approved the 802.11 standard, which laid down the specifications for WLAN (Wireless LAN). Subsequently, revisions were made to this standard and these resulted in three other standards—802.11a, 802.11b and 802.11g. The entire family of 802.11 standards are collectively called WiFi. The letters appended to the standard indicate the task group that proposed the modifications: for example, task group 'b' put

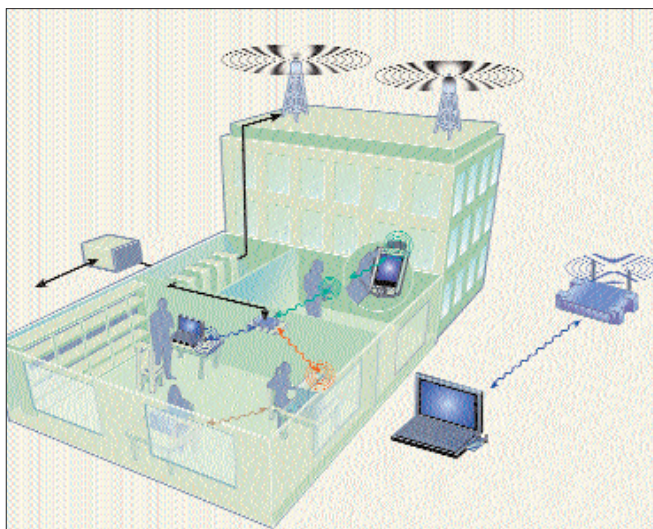
forth the 802.11b standard.

The name WiFi was first given to the 802.11b standard. Approved in 1999, 802.11b offered extensions to the original 802.11 that improved the highest data rate from 2 Mbps to 11 Mbps. It operated in the 2.4 GHz frequency range, like with the original, and had a range of about 300 feet. Range and speed are interrelated—as the wireless clients move further away from each other, the speed is bound to deteriorate.

802.11a operates at 5 GHz, and hence is incompatible with 802.11b. So, a laptop with an 802.11b card can't connect to an access point running on 802.11a. Also, the range of 802.11a is lower than that of 802.11b; hence, more access points are needed. On the bright side, it offers a much higher throughput—54 Mbps.

Interference is a problem with wireless networks. Bluetooth devices and cordless phones are some sources of this interference. 802.11a is less susceptible to interference compared to standards such as 802.11b.

To solve the limitations of the 'b' standard, and at the same time provide the speed of the 'a' standard, 802.11g was introduced. 802.11g offers speeds of 54 Mbps,



Here's a typical Wi-Fi setup in an office environment

Strip AD